

Shell project

Script to automate the process of listing all the resources in an AWS account.

Script :

```
#!/bin/bash

#####
#####
# Author: Shiv
# Version: v0.0.1

# Script to automate the process of listing all the resources in an AWS account
#
# Below are the services that are supported by this script:
# 1. EC2
# 2. RDS
# 3. S3
# 4. CloudFront
# 5. VPC
# 6. IAM
# 7. Route53
# 8. CloudWatch
# 9. CloudFormation
# 10. Lambda
# 11. SNS
# 12. SQS
# 13. DynamoDB
# 14. VPC
# 15. EBS
#
# The script will prompt the user to enter the AWS region and the service for
```

which the resources need to be listed.

```
#
# Usage: ./aws_resource_list.sh <aws_region> <aws_service>
# Example: ./aws_resource_list.sh us-east-1 ec2
#####
#####
```

```
# Check if the required number of arguments are passed
if [ $# -ne 2 ]; then
    echo "Usage: ./aws_resource_list.sh <aws_region> <aws_service>"
    echo "Example: ./aws_resource_list.sh us-east-1 ec2"
    exit 1
fi
```

```
# Assign the arguments to variables and convert the service to lowercase
aws_region=$1
aws_service=$2
```

```
# Check if the AWS CLI is installed
if ! command -v aws &> /dev/null; then
    echo "AWS CLI is not installed. Please install the AWS CLI and try again."
    exit 1
fi
```

```
# Check if the AWS CLI is configured
if [ ! -d ~/.aws ]; then
    echo "AWS CLI is not configured. Please configure the AWS CLI and try again."
    exit 1
fi
```

```
# List the resources based on the service
case $aws_service in
    ec2)
        echo "Listing EC2 Instances in $aws_region"
        aws ec2 describe-instances \
```

```

--region $aws_region \
--query "Reservations[*].Instances[*].InstanceId" \
--output text

;;
rds)
    echo "Listing RDS Instances in $aws_region"
    aws rds describe-db-instances --region $aws_region
    ;;
s3)
    echo "Listing S3 Buckets in $aws_region"
    aws s3api list-buckets --region $aws_region
    ;;
cloudfront)
    echo "Listing CloudFront Distributions in $aws_region"
    aws cloudfront list-distributions --region $aws_region
    ;;
vpc)
    echo "Listing VPCs in $aws_region"
    aws ec2 describe-vpcs --region $aws_region
    ;;
iam)
    echo "Listing IAM Users in $aws_region"
    aws iam list-users --region $aws_region
    ;;
route53)
    echo "Listing Route53 Hosted Zones in $aws_region"
    aws route53 list-hosted-zones --region $aws_region
    ;;
cloudwatch)
    echo "Listing CloudWatch Alarms in $aws_region"
    aws cloudwatch describe-alarms --region $aws_region
    ;;
cloudformation)
    echo "Listing CloudFormation Stacks in $aws_region"
    aws cloudformation describe-stacks --region $aws_region

```

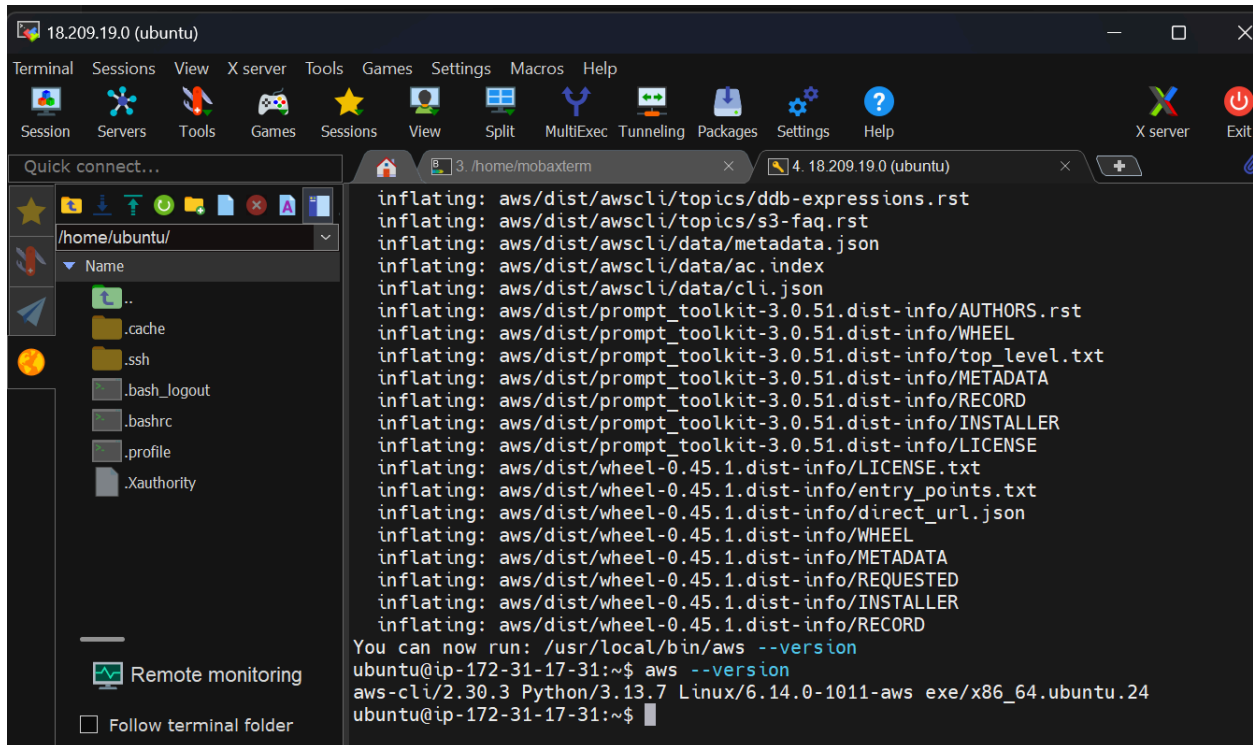
```

;;
lambda)
    echo "Listing Lambda Functions in $aws_region"
    aws lambda list-functions --region $aws_region
;;
sns)
    echo "Listing SNS Topics in $aws_region"
    aws sns list-topics --region $aws_region
;;
sqs)
    echo "Listing SQS Queues in $aws_region"
    aws sqs list-queues --region $aws_region
;;
dynamodb)
    echo "Listing DynamoDB Tables in $aws_region"
    aws dynamodb list-tables --region $aws_region
;;
ebs)
    echo "Listing EBS Volumes in $aws_region"
    aws ec2 describe-volumes --region $aws_region
;;
*)
    echo "Invalid service. Please enter a valid service."
    exit 1
;;
esac

```

Aws Cli installation on linux : [Installing or updating to the latest version of the AWS CLI - AWS Command Line Interface](#)

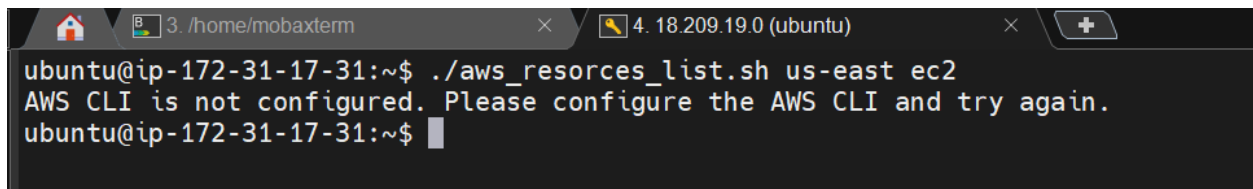
unzip awscliv2.zip → run this before curl command



The screenshot shows a Mobaxterm window with a terminal session on an Ubuntu 18.209.19.0 machine. The terminal displays the progress of installing the AWS CLI. It lists various files being inflated, including metadata, CLI JSON files, and prompt toolkit information. At the end, it provides the command to run the AWS CLI and shows the output of the version command.

```
inflating: aws/dist/awscli/topics/ddb-expressions.rst
inflating: aws/dist/awscli/topics/s3-faq.rst
inflating: aws/dist/awscli/data/metadata.json
inflating: aws/dist/awscli/data/ac.index
inflating: aws/dist/awscli/data/cli.json
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/AUTHORS.rst
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-17-31:~$ aws --version
aws-cli/2.30.3 Python/3.13.7 Linux/6.14.0-1011-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-17-31:~$
```

After Installation validation we get validation for configuration:



The screenshot shows a terminal window where the user has run the command `./aws_resorces_list.sh us-east ec2`. The output indicates that the AWS CLI is not configured and provides instructions to configure it.

```
ubuntu@ip-172-31-17-31:~$ ./aws_resorces_list.sh us-east ec2
AWS CLI is not configured. Please configure the AWS CLI and try again.
ubuntu@ip-172-31-17-31:~$
```

To resolve this go to aws account → security creddentials → create access key

aws configure command and provide access key

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key

 AKIAZVMTVJV7KF67JUC3

 ***** [Show](#)

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

```
ubuntu@ip-172-31-17-31:~$ aws configure
AWS Access Key ID [None]: AKIAZVMTVJV7KF67JUC3
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-17-31:~$
```

```
ubuntu@ip-172-31-17-31:~$ ./aws_resorces_list.sh us-east-1 ec2
Listing EC2 Instances in us-east-1
i-06aab5fbf5e7fea91
i-0acc87dae8fe297a7
i-031381a583aebfdbb
```

Done!